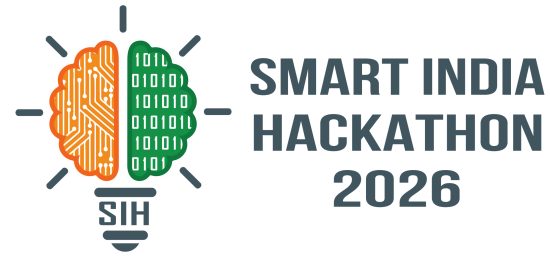


SMART INDIA HACKATHON 2026



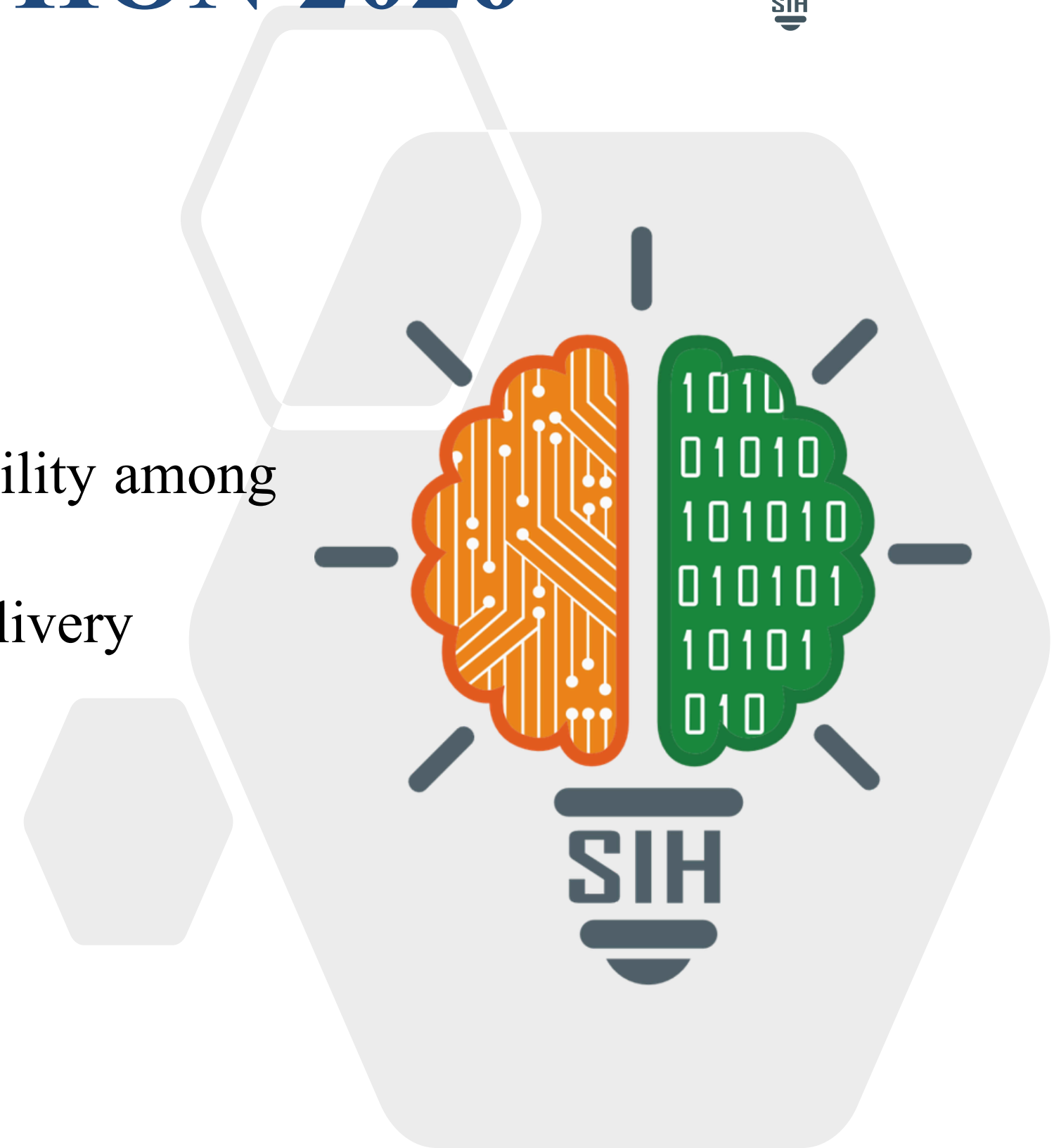
Problem Statement ID - SIH26129

Problem Statement Title - System integration and interoperability among government digital platforms, resulting in fragmented service delivery

Theme - Miscellaneous

PS Category – Software

Team Name – OneDesk



SOLUTION

A secure, standards-based interoperability platform that connects existing government portals, registries, and legacy systems into one citizen-facing experience - eliminating duplicate data entry, manual cross-department verification, and fragmented application status.

- **Federated Identity** - One OneDesk ID resolves multiple departmental identifiers (Revenue, Education, Welfare, Business Registry) into a single master citizen record, without departments losing ownership of their own data.
- **Consent-Based Data Exchange** - every cross-department data pull is purpose-bound, time-limited, and revocable by the citizen, modelled on India's DEPA / Account Aggregator consent-artifact standard.
- **Unified Application Tracking** - one application (e.g. Business License) is tracked end-to-end across Identity, Business Registry, Tax, and Municipal verification as a single status - not four separate portal login.
- **Legacy + Modern Connector Framework** - adapter layer speaks REST, Legacy SOAP, and Database/File connectors so existing department systems join without a rewrite.
- **Government Operations Console** - real-time visibility into connected systems, workflow orchestration, data quality, exceptions, SLA compliance, and a full audit trail for the officials.

ADDRESSAL APPROACH

Secure Data Exchange: A canonical data-mapping layer resolves field-level mismatches between department schemas (e.g. full-name/DOB vs name/dateOfBirth) before any record crosses a boundary, with schema validation and identifier resolution logged at every step.

Trust & Accountability: Purpose-bound consent tokens plus an immutable audit ledger - who accessed what, when, under which authorization - mean access without valid consent is blocked outright, not just logged after the fact.

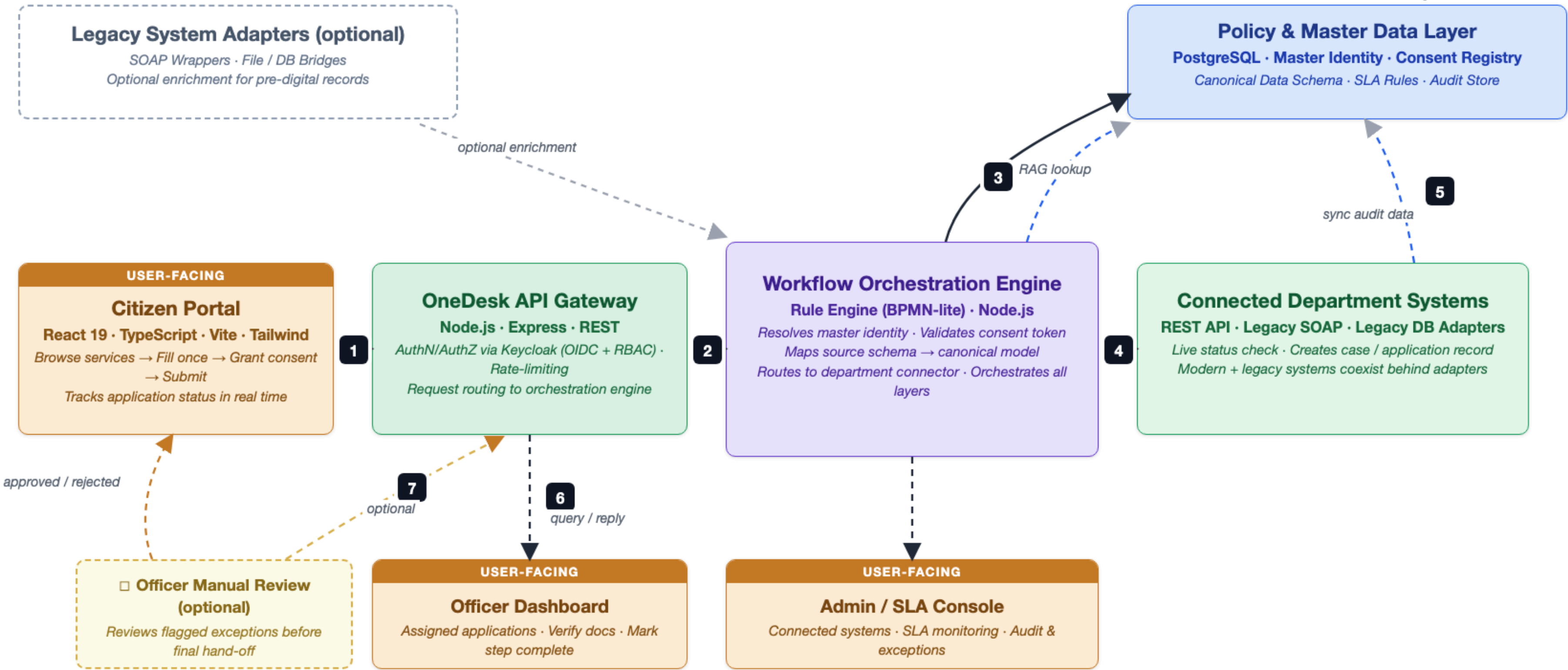
UNIQUE VALUE PROPOSITIONS

Zero Data Re-entry - citizens submit income, identity, and business details once; departments fetch verified records directly instead of asking for the same document again.

Graceful Degradation - if one department's system goes down mid-workflow, only that step pauses and auto- retries; the rest of the application keeps moving instead of failing outright.

Progressive Modernization - legacy systems join via adapters immediately and modernize on their own timeline (Integrate → Standardize → Modernize → Migrate → Retire) - no forced, disruptive replacement.

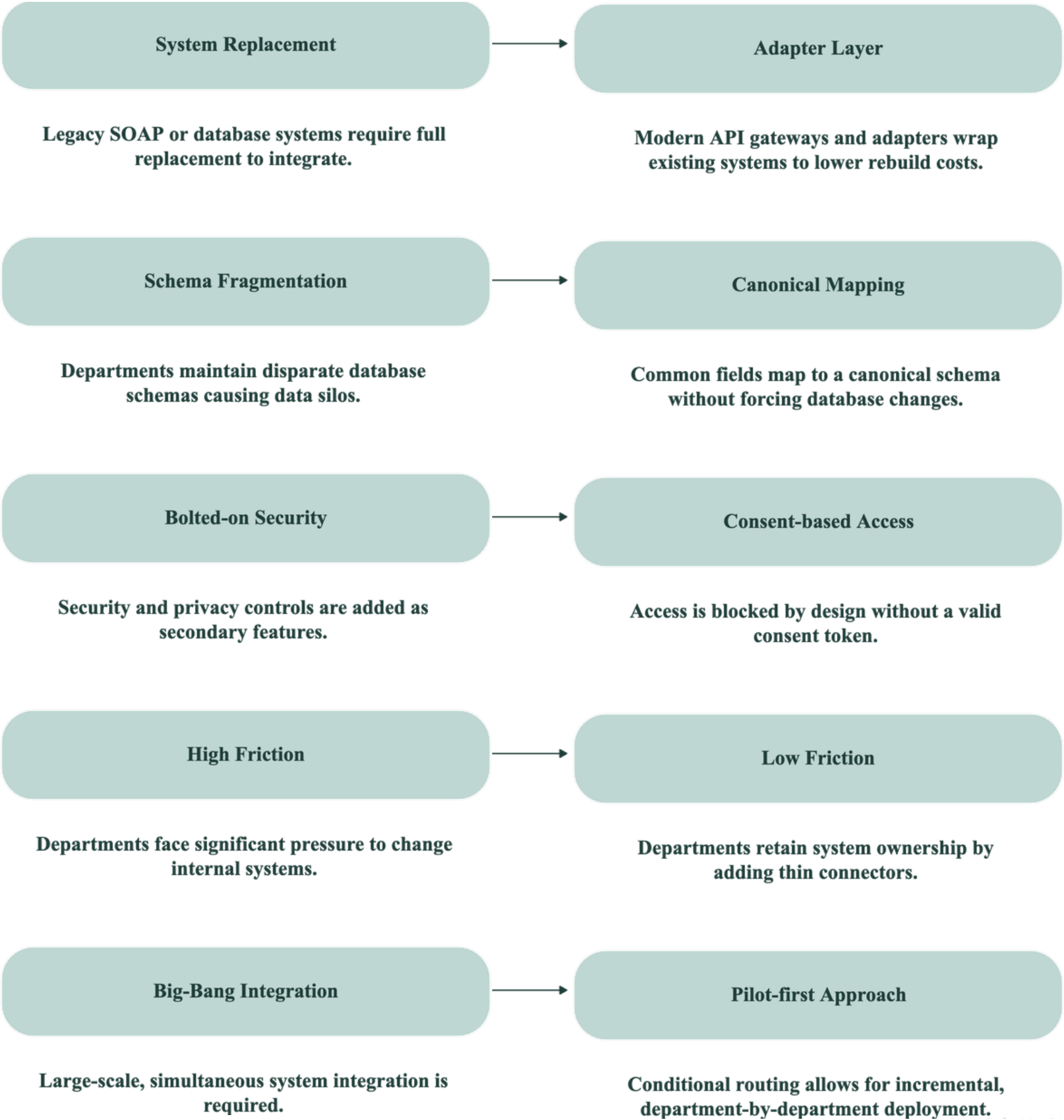
Full Traceability - every access, consent grant/revoke, and workflow event is timestamps and auditable, giving auditors and citizens the same evidence trail.



TECH STACK & COMPONENTS

- React 19
- TypeScript
- Node.js
- Express
- Tailwind CSS
- PostgreSQL
- React Router
- Radix UI
- Keycloak
- Git
- API REST / SOAP

IDEA FEASIBILITY



POTENTIAL CHALLENGES AND RISKS

- **Legacy & Data Fragmentation** - Schema drift, undocumented legacy systems, and inconsistent citizen records complicate integration and identity resolution.
- **Trust & Adoption** - Citizen consent friction and departmental ownership concerns can slow cross-department data sharing.
- **Security & Privacy** - A shared interoperability layer becomes a high-value target requiring zero-trust access, encryption, consent enforcement, and complete auditability.
- **Scalability & Resilience** - Government-scale traffic and centralized dependencies require horizontal scaling, redundancy, circuit breakers, and graceful degradation.

STRATEGIES FOR OVERCOMING CHALLENGES

- **Legacy & Data Quality** - Versioned adapter contracts, automated schema-validation tests, and Master Identity Resolution with fuzzy matching + manual review for mismatches.
- **Trust & Adoption** - Granular, plain-language consent with citizen review/revocation, combined with phased pilots that demonstrate measurable benefits before wider departmental rollout.
- **Security & Privacy** - Zero-trust connector architecture, mutual TLS, time-bound consent tokens, and complete audit logging for every cross-department data access.
- **Scalability & Resilience** - Async event processing, horizontal gateway scaling, SLA alerts, and per-connector circuit breakers with graceful degradation so one failed department does not block the entire platform.

POTENTIAL IMPACT ON TARGET AUDIENCE

CITIZENS

One login, one profile, apply once — no re-submitting same documents across depts. Real-time tracking replaces “visit office to check status” trips.

RURAL / LOW-LITERACY CITIZENS

Fewer physical office visits, fewer document copies needed = lower cost/time burden, especially for those far from dept offices.

DEPARTMENT OFFICERS

Single dashboard for assigned applications instead of juggling multiple legacy portals; verified data pre-fetched from other depts cuts manual cross-checking.

PLATFORM ADMINISTRATORS

Centralized SLA monitoring, exception handling, audit trail — visibility across depts instead of siloed reporting.

BUSINESSES / ENTREPRENEURS

Flagship Business License flow shows multi-dept approval (identity + registry + tax + municipal) in one tracked application, cuts licensing delay.

VULNERABLE GROUPS

Fewer in-person visits = fewer lost work-days, less dependency on middlemen/agents for form-filling.

BENEFITS OF THE SOLUTION

SOCIAL

Reduces citizen-official friction and corruption opportunity (fewer manual touchpoints, full audit trail). Builds trust via transparent tracking + consent control over own data.

ECONOMIC

Cuts processing time/cost per application (shared verified data vs duplicate manual checks per dept). Faster business licensing = faster local economic activity. Saves citizen income lost to repeated office visits.

GOVERNANCE / ADMINISTRATIVE

SLA monitoring surfaces bottlenecks objectively; exception/audit tabs enable accountability instead of blame-shifting between depts. Modernization tab gives clear legacy-to-modern migration roadmap, not forced rip-and-replace.

ENVIRONMENTAL

Fewer physical visits and paper submissions (digital consent, digital document reuse) cuts paper use and travel-related emissions at scale.

DATA PRIVACY

Consent-first architecture gives citizen explicit control per data category — improves data protection standards across govt versus current ad hoc practice.

SCALABILITY / FUTURE-PROOFING

Interoperability Layer pattern extends to new depts/services without rearchitecting — long-term platform, not one-off portal.

1

PLATFORM

Unified access for all government services

40-60%

Reduction in processing time

70% LESS

Paper usage and physical visits

HIGHER

Transparency, accountability & trust

CITIZEN-CENTRIC

Secure, consent-driven data architecture

REFERENCES

- India API Setu — Govt. Interoperability Platform (MeitY) - [link](#)
- India Enterprise Architecture Framework (IndEA) 1.0 - [link](#)
- European Interoperability Framework (EIF) - [link](#)
- OAuth 2.0 Authorization Framework - RFC 6749 - [link](#)
- OpenID Connect Core 1.0 - [link](#)
- Canonical Data Model - Enterprise Integration Patterns - [link](#)
- API Gateway Pattern microservices.io - [link](#)
- X-Road Data Exchange Layer, Estonia - [link](#)

JOURNALS AND PAPERS

- Sarantis et al. - The Greek Electronic Government Interoperability Framework: Standards and Infrastructures for One-Stop Service Provision (IEEE) - [link](#)
- E-government Data Interoperability Framework in Hong Kong (IEEE) - [link](#)
- Advancing the One-Stop-Shop e-Government Paradigm (IEEE) - [link](#)
- Trustworthy Cross-Border Interoperable Identity System for Developing Countries (arXiv, 2023) - [link](#)

GitHub : [One Desk - System Integrations and Interoperability among Govt Orgs](#)

THANK YOU

